Group: Jason Antunes, [redacted]

Data Management Artifacts

Data Dictionary

Field Name	Data Type	Data Format	Nulls OK?	Description	Reason for inclusion
Client	Text	"Client" + capital last letter of the alphabet	No	Anonymous name given to the consultation client	Primary information
Subclient	Text	Typically Initials or generic description	Yes	When a client organization includes one or one consultations	Part of the original data provided
Weekday	Text	Three letter abbreviation of the day of the week	No	formulated from date with = TEXT([Date], "ddd")	Part of the original data provided, but we separated the day of the week for faster analysis of weekday and date
Date	Date	dd-mm-yyyy	No	Date the consultation took place	Primary information
Start	Time	hh:mm	No	Start time of the meeting.	Primary information
End	Time	hh:mm	No	End time of the meeting.	Primary information
Total time	Time	(hh):mm	No	Duration of the meeting. Formulated with =MOD([End]-[Start], 1)	Provided in the original data set
Hourly rate	Currency	$XX.XX	No	Consultation fee for 60 minutes	Primary information
Charge	Currency	$XX.XX	No	Formulated from hourly rate and total time. Formulated with =[Total time]*24*[Hourly Rate]	Provided in the original data set
Charge notes	Text	Short note	Yes	Notes on discounts, if it has been paid, and note on foreign currency used	Provided in the original data set
Invoice Date	Date	dd-mm-yyyy	Yes	Date an invoice was sent to the client	Provided in the original data set
Invoice Amount	Currency	$XX.XX	Yes	Amount provided for the invoice.	Provided in the original data set
Notes	Text	Longer note	Yes	Notes from the consultant about what was discussed in the meeting	Provided in the original data set

A.copy.of.the.data.dictionary.in.;xlsx.format.is.also.available.and.has.been.submitted.alongside.this.report;

Data Lineage Documentation

The dataset we received was an Excel file with 7 worksheets. Most of the worksheets had the same columns and formats, though there were some slight variations. **Text in bold (aside from section titles) indicates changes made to the entries.**

Worksheets Overall

- Date column format: Custom > dddd", "mmmm" "d", "yyyy
  - (eg: Thursday, November 27, 2025)
  - **Changing to dd-mm-yyyy**
  - **Adding a secondary column to hold the day of the week information**
- Start and End columns format: Custom > hh:mm:ss (24 hour format)
  - **Changing to hh:mm since all of the :ss are zeros**
- Hourly Rate and Charge columns: formatted as text with $.
  - **Changing to numeric**
- Notes Column formatted with word wrap and right alignment.
  - **Changing to Align Left and un-wrap text**

Worksheet 1: Outstanding Payments

- 8 columns, 0 rows + header row
- Column names: Date, Client, Start, End, (Mins) Hourly Rate, Charge, Notes

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Data Management Artifacts

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Notes	Text	Longer note	Yes	Notes from the consultant about what was discussed in the meeting	Provided in the original data set

A copy of the data dictionary in .xlsx format is also available and has been submitted alongside this report.

Data Lineage Documentation

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Worksheet 1: Outstanding Payments

- 8 columns, 0 rows + header row
- Column names: Date, Client, Start, End, (Mins) Hourly Rate, Charge, Notes

Group: Jason Antunes, [redacted]

Notes:

- Empty page

Worksheet 2: Nov. 27 -- Client A (not paid)

- 8 columns, 18 rows + header row
- Column names: Date, Client, Start, End, Total Time (Mins), Hourly Rate, Charge, Notes

Notes:

- Only 15 entries in 18 rows.
- Looks like a couple rows hold monthly totals for November and December.
    - Noting them here and then removing
    - November: Time: 189, Charge: $126.00
    - December: Time: 340, Charge: $453.33
- Rows 17 & 18 have a different format (Custom > hh:mm)

Worksheet 3: Oct. 26-Nov. 20 -- Client A (pa

"Client A (pa" is as shown in the MS Excel worksheet. This was possibly a typo and was intended to say "(paid)"; however, that is an assumption.

- 7 Columns, 20 rows + header row
- Column names: Date, Client, Start, End, Total Time (Mins), Hourly Rate, Notes

Notes:

- Only 19 entries in 20 rows. 1 row holds totals of Total Time (Min), Hourly Rate
    - Noting here then removing
    - Total Time: 1318, Hourly Rate: $878.67

Worksheet 4: Sept. 22-Oct. 12 -- Client A

- 9 Columns, 17 rows + header row
- Column names: Date, Client, Start, End, Total Time (Mins), Hourly Rate, Charge, Payment

Notes:

- 1 empty row (9) **Deleting to change table to 16 rows + header row**
- 1 row with totals for Total Time (Mins) and Charge
    - Noting here and then deleting
    - Total Time: 1019, Charge: $1406.33
- Date format is different. Custom > mmmm "d", yyyy
    - **Changing to standard**
- Hourly Rate column formats: 7 as General, 3 as Currency, 5 as Text
    - **Changed all to numeric**
- Charge Column: mostly empty. Only has entries at row 2 and row 17, and a totals row.
    - **Changed to numeric**
- Payment Column: Empty **Deleting this column since it isn't used**
- Notes Column: 8 are empty and we have no way of filling them

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Worksheet 5: Payment LinksInvoices Sent

- Messy with data misaligned with the column titles and data formats aren't standardized.
- 10 columns, 138 rows + header row
- Column names: Date, Client, Start, End, (Mins), Rate, Notes, Link/Invoice Sent, Amount, USD

Notes:

- 1 empty row. **Deleting**
- 1 incomplete row. **Deleting because it is an outlier.**
- Date Column: 3 are in d-m-yyyy, and 75 don't have the day of the week.
    - 2 rows are empty except for a note for H. **Deleting as it is not relevant**
- Client: top 20 rows look like they were formatted as a drop down to include names C. and T. **Removing the drop down to get remove reference to these names.**

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Worksheet 6: Clients with Issues

- 8 columns, 1 row + header row
- Column Names: Date, Client, Subclient, Start, End, Total Time (Mins), Hourly Rate, Notes

Worksheet 7: Free Consults

- 7 columns, 2 rows + header row
- Column Names: Date, Client, Start, End, Total Time (Mins), Hourly Rate, Notes

Bringing the Data Together:

Now we will bring all the data from the separate worksheets into one single Excel file for analytics: Consulting.xlsx.

Each worksheet had slightly different columns used, so the following table shows what columns were used in which sheet. The highlighted cells show the columns that will be used in the analytics file. Additionally, a column for Weekday was added.

1 Outstanding Payments	Date	Client		Start	End	(Mins)	Hourly Rate	Charge		Notes			
2 Nov.27 -	Date	Client		Start	End	Total Time (Mins)	Hourly Rate	Charge		Notes			
3 Oct. 26- Nov. 20	Date	Client		Start	End	Total Time (Mins)	Hourly Rate			Notes			
4 Sept. 22-Oct. 12	Date	Client		Start	End	Total Time (Mins)	Hourly Rate	Charge	Payment	Notes			
5 Payment Links/Invoices	Date	Client		Start	End	(Mins)	Rate			Notes	Link/Invoice Sent	Amount	USD
6 Clients with Issues	Date	Client	Subclient	Start	End	Total Time (Mins)	Hourly Rate			Notes			
7 Free Consults	Date	Client		Start	End	Total Time (Mins)	Hourly Rate			Notes			

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Data Quality Assessment Report

Duplicates Removed

Client	Weekday	Date	Start	End	Total Time	Hourly Rate	Charge - Calculated	Notes
Client A	Mon	22-09-2025	22:01	00:24	02:23	$100.00	$ 238.33	Advised on publishing and income generation
Client A	Mon	22-09-2025	11:18	13:29	02:11	$ 80.00	$ 174.67	Advised on publishing
Client A	Mon	22-09-2025	14:18	14:34	00:16	$ 80.00	$ 21.33	Advised on publishing
Client A	Fri	26-09-2025	12:58	01:14	00:16	$ 80.00	$ 21.33	Advised on workshop and podcast idea
Client A	Sun	05-10-2025	12:07	13:58	01:51	$ 80.00	$ 148.00	
Client A	Sun	05-10-2025	18:47	19:12	00:25	$ 80.00	$ 33.33	
Client A	Sun	05-10-2025	20:04	21:03	00:59	$ 80.00	$ 78.67	
Client A	Sun	05-10-2025	12:07	13:58	01:51	$ 80.00	$ 148.00	Advised on publishing brand
Client A	Sun	05-10-2025	18:47	19:12	00:25	$ 80.00	$ 33.33	Advised on publishing brand
Client A	Sun	05-10-2025	20:04	21:03	00:59	$ 80.00	$ 78.67	Advised on publishing brand
Client A	Sun	05-10-2025	12:07	13:58	01:51	$ 80.00	$ 148.00	
Client A	Sun	05-10-2025	18:47	19:12	00:25	$ 80.00	$ 33.33	
Client A	Sun	05-10-2025	20:04	21:03	00:59	$ 80.00	$ 78.67	
Client A	Thu	09-10-2025	13:39	15:06	01:27	$ 80.00	$ 116.00	
Client A	Thu	09-10-2025	19:40	22:06	02:26	$ 80.00	$ 194.67	
Client A	Fri	10-10-2025	13:00	13:15	00:15	$ 80.00	$ 20.00	
Client A	Fri	10-10-2025	13:17	13:39	00:22	$ 80.00	$ 29.33	
Client A	Sun	12-10-2025	12:37	13:30	00:53	$ 80.00	$ 70.67	

Outliers Taken Out:

Client	Subclient	Weekday	Date	Start	End	Total Time	Time (Mins)	Hourly Rate
Client F								$ 600.00

"Note: Programming and coordinating each session (idea, writing description, inviting guests, asking for bios)"

This hourly rate is a one-time charge and is more than twice the next highest hourly rate. Deleting as an outlier. It was also an incomplete row, which is another reason to remove it.

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Empty Values Filled In

Client	Subclient	Weekday	Date	Start	End	Total Time	Hourly Rate
Client C		Thu	06-11-2025	13:00	14:00	01:00	$147.50
Client C		Thu	06-11-2025	14:00	15:03	01:03	$147.50
This is the hourly rate used for Client C in the other entries							
Client K		Sun	30-11-2025	10:48	11:48	01:00	$150.00
This is the hourly rate used for Client K							
Client D		Tue	04-11-2025	13:00	13:56	00:56	$225.00
Client D		Tue	04-11-2025	14:00	15:29	01:29	$225.00
This is the most common hourly rate used for Client D. The only other rate is marked with a promo							
Client F	DB	Fri	31-10-2025	12:00	12:30	00:30	$150.00
Client F	KP	Thu	13-11-2025	11:10	11:40	00:30	$150.00
Client F	ML	Fri	14-11-2025	12:00	12:30	00:30	$150.00
Client F	Teaching	Sat	22-11-2025	12:00	14:00	02:00	$150.00
Client F	JL	Sun	23-11-2025	14:00	14:30	00:30	$150.00
30 minutes is the most common duration, and 12:00 is the most common start time on Fridays for this Client. The hourly rates are the standard for this client.							

Possible Errors or Miscalculations

Client	Subclient	Weekday	Date	Start	End	Total Time	Time (Mins) - Given	Hourly Rate	Charge - Calculated	Charge - given
Client A		Sun	12-10-2025	12:37	13:30	00:53	53	$ 80.00	$ 70.67	$1,168

Client	Subclient	Weekday	Date	Start	End	Total Time	Time (Mins) - Given	Hourly Rate	Charge - Calculated	Charge - given
Client C		Thu	06-11-2025	13:00	14:00	01:00	73.75	$ 147.50	$ 147.50	
Client C		Thu	06-11-2025	14:00	15:03	01:03	73.75	$ 147.50	$ 154.88	
Client D		Tue	04-11-2025	13:00	13:56	00:56	62.5	$ 225.00	$ 210.00	
Client D		Tue	04-11-2025	14:00	15:29	01:29	62.5	$ 225.00	$ 333.75	

The highlighted data is from our source dataset. But when total time or total charge was calculated through Excel, these discrepancies showed up.

There may be individual reasons as to why these are different from the calculations, but for the purposes of our work, we will be removing them and using the calculated amounts for data analysis.

Group: Jason Antunes, [redacted]

Review of Cleaned Dataset:

	Client	Subclient	Weekday	Date	Start	End	Total Time
Count	168	23	168	168	168	168	168
Average							01:06
Minimum				21-08-2025			00:02
Maximum				18-01-2026			04:58
Range				150			04:56

	Hourly Rate	Charge	Charge Notes	Invoice Date	Invoive Amount	USD	Notes
Count	168	168	48	71	12	12	139
Average	$117.63	$ 138.37			918.43		
Minimum	$ 40.00	$ 1.33		04-09-2025	132.75		
Maximum	$225.00	$1,012.50		29-11-2025	7,552.50		
Range	$185.00	$1,011.17		86	7,419.75		

Our data dictionary says the highlighted columns cannot have null values, and none of them do.

We can see the dates span from late August 2025 to mid January 2026; therefore, the information is fairly recent.

Limitations and Impact on Analysis:

- This data set is extremely small with less than 200 entries. Even with k-fold cross validation methods, any machine learning algorithms will be limited in their capabilities.
- Currently the dataset is set up to look at each meeting. However, there are often multiple meetings with the same client on the same day. If we aggregate the data by day, the data set will become even smaller. The following snippet shows 7 meetings, but they are over only 3 days.

Client	Weekday	Date	Start	End	Total Time	Hourly Rate	Charge
Client A	Mon	03-11-2025	12:40	12:48	00:08	$ 40.00	$ 5.33
Client A	Mon	03-11-2025	14:18	14:58	00:40	$ 40.00	$ 26.67
Client A	Fri	07-11-2025	08:01	12:00	03:59	$ 40.00	$ 159.33
Client A	Fri	07-11-2025	12:30	14:00	01:30	$ 40.00	$ 60.00
Client A	Fri	07-11-2025	14:20	15:29	01:09	$ 40.00	$ 46.00
Client A	Fri	07-11-2025	22:32	23:12	00:40	$ 40.00	$ 26.67
Client A	Sat	08-11-2025	11:18	13:19	02:01	$ 40.00	$ 80.67

- The data spans a date range just shy of 6 months, which limits the accuracy potential with exploration via time-series calculations.
- While we have data on invoices, there may not be enough to provide any useful information on billing cycles or how prompt clients are with their payments.

Resources

Harvard College, (© 2026). *Data Dictionary*. *https://datamanagement.hms.harvard.edu/collect-analyze/documentation-metadata/data-dictionary*